Maxwell's eqns. for Magnetostatics and Electrostatics

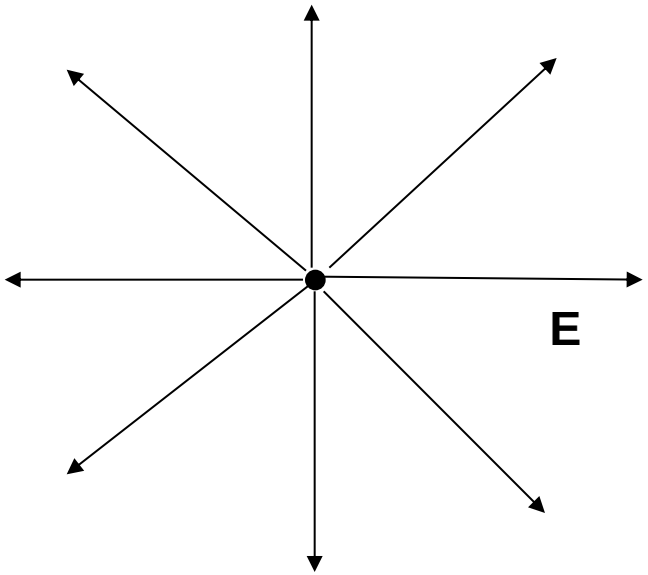
$$\vec{\nabla} \cdot \vec{E} = \frac{1}{\varepsilon_0} \rho$$ **Gauss's law**

$$\vec{\nabla} \times \vec{E} = 0$$

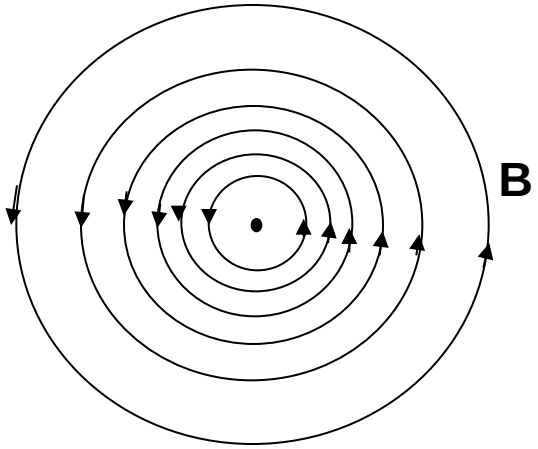
$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

Ampere's law



Electrostatic field of a point charge



Magnetostatic field of a long wire

Electric field diverges from a point – Magnetic fields curl around a current

Electric field lines originate on a +ve charge and end on a –ve charge

Magnetic fields do not begin or end anywhere – they just form closed loops

There are no point sources for B, as there are for E, as proved by div B = 0

There is no “magnetic monopole” and no magnetic version of Coulomb’s law

As Ampere put it, magnetic fields are produced by electric charges in motion, and it takes another moving charge to “feel” the magnetic field

Magnetic vector potential

In electrostatics, we know $\vec{\nabla} \times \vec{E} = 0$

Hence we had a scalar potential $\vec{E} = -\nabla V$ since curl grad is zero

Similarly, in magnetostatics we have $\vec{\nabla} \cdot \vec{B} = 0$

So we can write B as the curl of some vector **A**

$$\vec{B}(\vec{r}) = \vec{\nabla} \times \vec{A}(\vec{r})$$

since Divergence of a curl is always zero

$\vec{A}(\vec{r})$ **is the magnetic vector potential**

Unit of $\vec{A}$ will be volt-second / meter

We know that magnetic induction vector **B** according to Biot-Savart's law is given by

$$\mathbf{B}=\frac{\mu_0}{4\pi}\int I\,\frac{d\mathbf{l}\times\mathbf{r}}{r^3}$$

i.e.
$$\mathbf{B}=\frac{\mu_0 I}{4\pi}\int d\mathbf{l}\times\left\{-\operatorname{grad}\left(\frac{1}{r}\right)\right\} \qquad \left[\text{as } \nabla\left(\frac{1}{r}\right)=\frac{-\mathbf{r}}{r^3}\right]$$

$$\operatorname{Curl} S\mathbf{A}=S\operatorname{curl}\mathbf{A}-\mathbf{A}\times\operatorname{grad} S$$

i.e.
$$-\mathbf{A}\times\operatorname{grad} S=\operatorname{curl} S\mathbf{A}-S\operatorname{curl}\mathbf{A}$$

So
$$\mathbf{B}=\frac{\mu_0 I}{4\pi}\int\left[\operatorname{curl}\frac{d\mathbf{l}}{r}-\frac{1}{r}\operatorname{curl}(d\mathbf{l})\right]$$

The second term in the integral is zero because $d\mathbf{l}$ does not depend on the co-ordinates of the field point while the first term on changing the order of differentiation and integration yields

$$\mathbf{B} = \text{curl} \int \frac{\mu_0 I \, d\mathbf{l}}{4\pi r}$$

or $$\mathbf{B} = \text{curl } \mathbf{A}. \qquad \ldots(A)$$

Thus we see that magnetic induction vector **B** can be represented as the curl of a vector function **A** given by

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{I \, d\mathbf{l}}{r} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J} \, d\tau}{r} \quad (\text{as } I \, d\mathbf{l} = \mathbf{J} \, d\tau). \qquad \ldots(B)$$

The vector function **A** is called magnetic vector potential.

The vector potential produced by line (filamentary), surface and volume currents are written as

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \oint_{line} \frac{\vec{I}(\vec{r}')dl'}{R}$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \oint_{a'} \frac{\vec{K}(\vec{r}')da'}{R}$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \oint_{V'} \frac{\vec{J}(\vec{r}')d\tau'}{R}$$

Regarding magnetic vector potential **A** it must be noted that—

(*i*) *It satifies Poisson's equation* :

By definition, as

$$\mathbf{A}=\frac{\mu_0}{4\pi}\int\frac{\mathbf{J}'}{r}\,d\tau'$$

$$\nabla^2\mathbf{A}=\frac{\mu_0}{4\pi}\int\nabla^2\left(\frac{\mathbf{J}'}{r}\right)d\tau'=\frac{\mu_0}{4\pi}\int\mathbf{J}'\,\nabla^2\left(\frac{1}{r}\right)d\tau'$$

(as $\mathbf{J}'$ is not a function of x, y, and z)

or $$\nabla^2\mathbf{A}=-\mu_0\,\mathbf{J}' \qquad \left[\text{as } \int\nabla^2\,(1/r)\,d\tau'=-4\pi\right]$$

(ii) *The line integral of magnetic vector* **A** *round a closed path gives the magnetic flux linked with the area enclosed by the closed path i.e.*

$$\Phi_B = \oint \mathbf{A} \cdot d\mathbf{l}$$

By definition

$$\Phi_B = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S \text{curl } \mathbf{A} \cdot d\mathbf{s} \qquad [\text{as } \mathbf{B} = \text{curl } \mathbf{A}]$$

i.e., $$\Phi_B = \oint \mathbf{A} \cdot d\mathbf{l} \qquad \left[\text{as } \int \text{curl } \mathbf{A} \cdot d\mathbf{s} = \oint \mathbf{A} \cdot d\mathbf{l}\right]$$

(*iii*) *The divergence of magnetic vector potential* A *is zero or a scalar constant.*

By definition of A

$$\mathbf{B} = \text{curl } \mathbf{A}$$

so $$\nabla \times \mathbf{B} = \nabla \times \nabla \times \mathbf{A} = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$$

But as

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad \text{and} \quad \nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$$

$$\nabla (\nabla \cdot \mathbf{A}) = 0 \quad i.e. \quad \text{grad div } \mathbf{A} = 0$$

This means that

i.e. div A is zero or forms a uniform scalar field with constant value.*

We can add to magnetic potential the gradient of any scalar function whose curl vanishes, with no effect on **B**

We use this freedom to eliminate the divergence of $\vec{A}$ and make it zero

$$\vec{\nabla} \bullet \vec{A} = 0$$

Suppose our original potential A_0 has a non-zero divergence. We can add to it the gradient of some function

$$\vec{A} = \vec{A}_0 + \vec{\nabla}\lambda \qquad \vec{\nabla} \cdot \vec{A} = \vec{\nabla} \cdot \vec{A}_0 + \nabla^2 \lambda$$

$$\nabla^2 \lambda = -\vec{\nabla} \cdot \vec{A}_0$$

This is identical to Poisson's equation $\nabla^2 V = -\frac{\rho}{\varepsilon_0}$

It is always possible to make the vector potential divergenceless

This is the non-uniqueness of the vector potential

example

Vector potential of an infinite solenoid with n turns per unit length, radius R and current I

We cannot use $$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \oint_{C'} \frac{I d\vec{l}\,'}{R}$$

since the current itself extends to infinity

Alternately we can go as follows

$$\oint \vec{A} \cdot d\vec{l} = \int \left(\vec{\nabla} \times \vec{A}\right) \cdot d\vec{a} = \int \vec{B} \cdot d\vec{a} = \phi$$ (magnetic flux)

This is similar to Ampere's law $$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$$

If symmetry permits, we can determine A from flux.

(Similar to the ampere's law problem of a fat wire carrying a uniform distributed current)

Vector potential is circumferential --- using Circular amperian loop of radius s inside the solenoid (note that the field outside is zero)

$$\oint \vec{A}\cdot d\vec{l} = A(2\pi s) = \int \vec{B}\cdot d\vec{a} = \mu_0 nI(\pi s^2)$$

$$\vec{A} = \frac{\mu_o nI}{2} s\hat{e}_\phi \quad \text{for s < R}$$

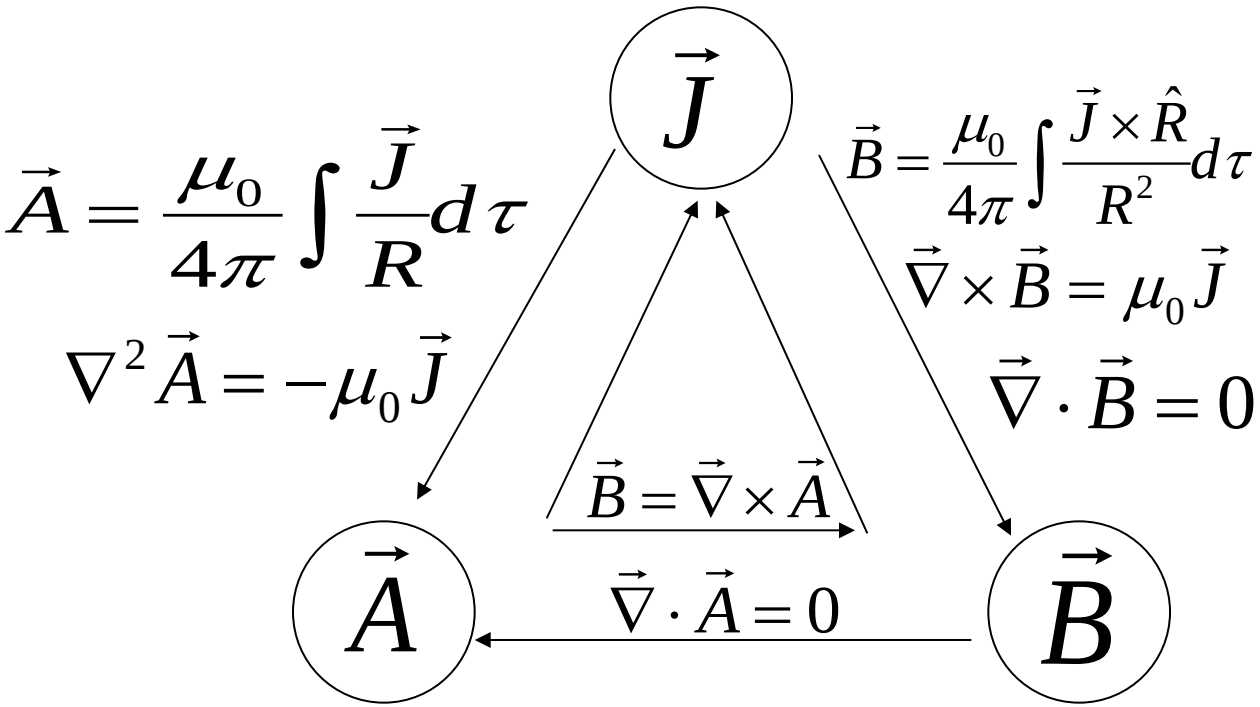
For an amperian loop outside the solenoid s > R, the flux is

$$\int \vec{B}\cdot d\vec{a} = \mu_0 nI(\pi R^2) \quad \text{since the field extends only up to R}$$

$$\vec{A} = \frac{\mu_o nI}{2}\frac{R^2}{s}\hat{e}_\phi \quad \text{for s > R}$$

Check whether $\vec{B} = \vec{\nabla}\times\vec{A}$ $\qquad \vec{\nabla}\bullet\vec{A} = 0$

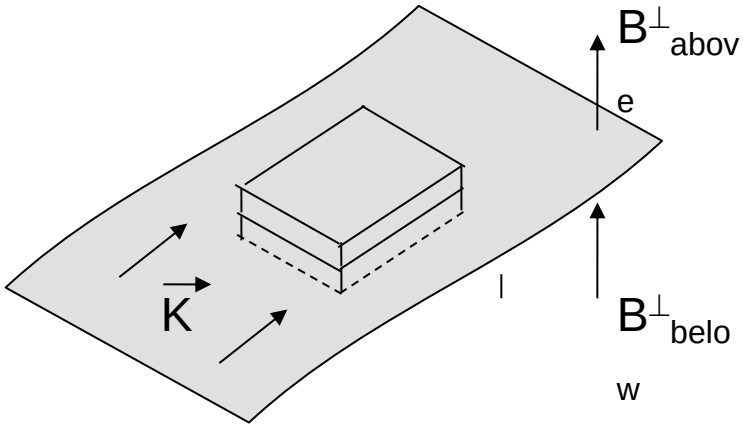
Summary of Magnetostatics in free space



Magnetostatic Boundary Condition

$\vec{\nabla} \cdot \vec{B} = 0$ Applying this in the integral form, ie,

$\oint \vec{B} \cdot \overrightarrow{da} = 0$ to a pill box

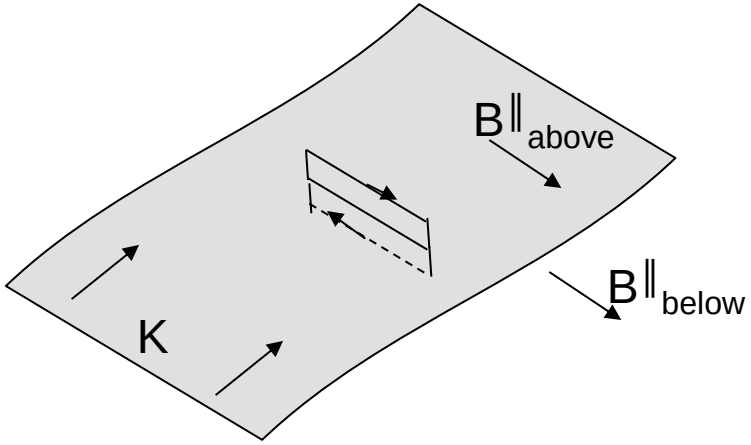


$$B^{\perp}_{above} = B^{\perp}_{below}$$

For tangential components
(perpendicular to the current)

$$\oint \vec{B} \cdot \vec{dl} = (B''_{above} - B''_{below})l = \mu_0 I_{enc} = \mu_0 Kl$$

$$B''_{above} - B''_{below} = \mu_0 K$$



For amperian loop running parallel to
current The parallel components are continuous.
Summarizing, we can write

$$\vec{B}_{above} - \vec{B}_{below} = \mu_0 (\vec{K} \times \hat{n})$$

n^ Unit vector perpendicular
to the surface

In Electrostatics, the scalar potential is continuous

Similarly, here, the magnetic vector potential is continuous

$$\vec{A}_{above} = \vec{A}_{below}$$

$$\nabla \cdot \vec{A} = 0$$

Guarantees that the normal component is continuous

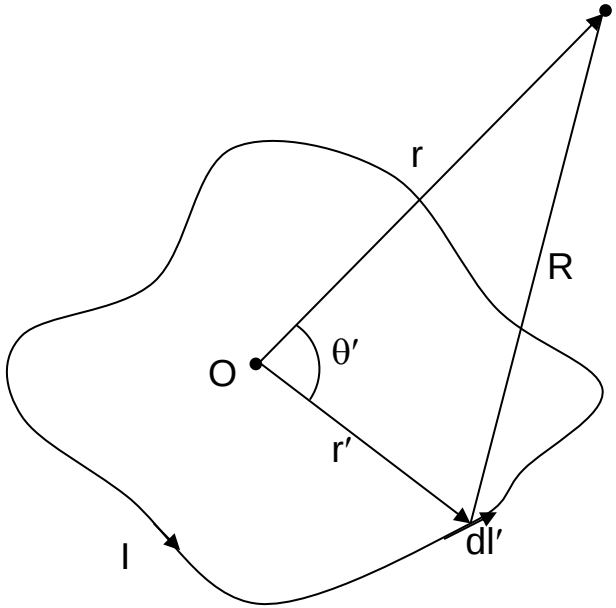
$$\nabla \times \vec{A} = \vec{B}$$

in the form

$$\oint \vec{A} \cdot \vec{dl} = \oint \vec{B} \cdot \vec{da} = \Phi$$

means that the tangential components are continuous (since the flux through an amperian loop of vanishing thickness is zero)

Multipole Expansion of the Vector Potential for line currents



We have seen that 1/R can be written as

$$\frac{1}{R} = \frac{1}{\sqrt{r^2 + (r')^2 - 2rr'\cos\theta'}} = \frac{1}{r}\sum_{n=0}^{\infty}\left(\frac{r'}{r}\right)^n P_n(\cos\theta')$$

Accordingly the vector potential of the current loop can be written as

$$\vec{A}(\vec{r}) = \frac{\mu_0 I}{4\pi} \oint \frac{1}{R} \overrightarrow{dl'} = \frac{\mu_0 I}{4\pi} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \oint (r')^n P_n(\cos\theta') \overrightarrow{dl'}$$

$$\vec{A}(\vec{r}) = \frac{\mu_0 I}{4\pi} \left[\frac{1}{r} \oint \overrightarrow{dl'} + \frac{1}{r^2} \oint r' \cos\theta' \overrightarrow{dl'} + \frac{1}{r^3} \oint (r')^2 \left(\frac{3}{2} \cos^2 \theta' - \frac{1}{2} \right) \overrightarrow{dl'} + \cdots \right]$$

$$\oint \overrightarrow{dl'} = 0$$ Total vector displacement around a closed loop is zero

$\Rightarrow$ **The magnetic monopole always vanishes**.

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0 I}{4\pi r^2} \oint r' \cos\theta' \overrightarrow{dl'} = \frac{\mu_0 I}{4\pi r^2} \oint (\hat{r} \cdot \vec{r}') \overrightarrow{dl'}$$

We can show that $$\oint (\hat{r} \cdot \vec{r}')\, \overrightarrow{dl'} = -\hat{r} \times \int \overrightarrow{da'}$$

Do this as homework

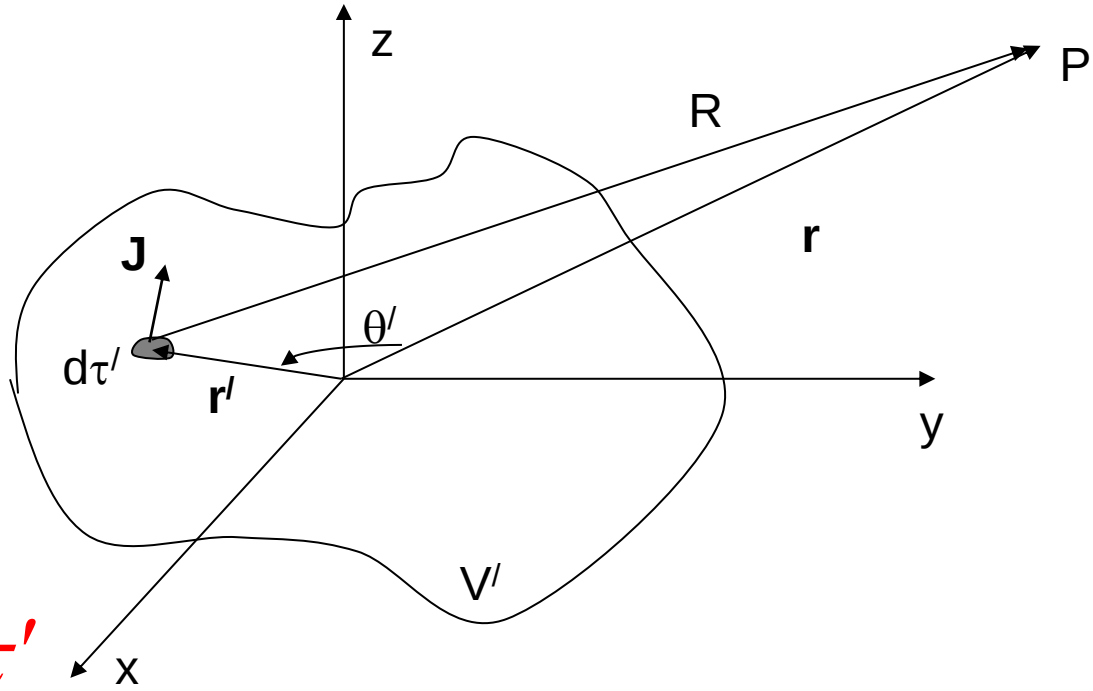
Hence $$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$

Where m is the Magnetic dipole moment given by

$$\vec{m} \equiv I \int \overrightarrow{da} = I\vec{a}$$

Multipole expansion of Vector potential for volume currents

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \oint_{V'} \frac{\vec{J}(\vec{r}\,')d\tau'}{R}$$



For your memory, I will be putting corresponding values in electrostatics in red color

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \oint_{V'} \frac{\rho(\vec{r}\,')d\tau'}{R}$$

$$R_i = (r^2 + r'^2 - 2rr'\cos\theta')^{1/2}$$

We take the condition r > r/ at all the time

$$\vec{A}(\vec{r}) = \vec{A}_M(\vec{r}) + \vec{A}_D(\vec{r}) + \vec{A}_Q(\vec{r}) + \ldots\ldots$$

$$= \frac{\mu_0}{4\pi}\int_{V'} \vec{J}(\vec{r}')d\tau' + \frac{\mu_0}{4\pi r^2}\int_{V'} \vec{J}(\vec{r}')(\hat{r}\cdot\vec{r}')d\tau' +$$

$$\frac{\mu_0}{4\pi r^3}\int_{V'} \frac{1}{2}\vec{J}(\vec{r}')\left[3(\hat{r}\cdot\vec{r}')^2 - r'^2\right]d\tau' + \ldots\ldots$$

$$V = \frac{1}{4\pi\varepsilon_0}\int_{V'} \rho(\vec{r}')d\tau' + \frac{1}{4\pi\varepsilon_0 r^2}\int_{V'} \rho(\vec{r}')r' \cos\theta' d\tau' +$$

$$\frac{1}{4\pi\varepsilon_0 r^3}\int_{V'} \frac{1}{2}\rho(\vec{r}')r'^2\left[3(\cos\theta')^2 - 1\right]d\tau'$$

Magnetic monopole term is always zero since

$$\oint_{V'} \vec{J}(\vec{r}')d\tau' = \sum_j I_j \oint_{C_j} d\vec{l}_j = 0$$

Total vector displacement around a closed loop is zero.. No magnetic monopoles in nature.. So, the dominant term is always the Dipole term

Dipole contribution to vector potential due to volume currents **For line currents**

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi r^2}\int_{V'} \vec{J}(\vec{r}\,')(\hat{r}\cdot\vec{r}\,')d\tau' \qquad \frac{\mu_0}{4\pi r^2}\int_{V'} I(\hat{r}\cdot\vec{r}\,')d\vec{l}\,'$$

Filamentary currents $\vec{J}d\tau$ **will be replaced by** $Id\vec{l}\,'$

Now using the identity (problem 1.61 in Griffiths)

$$\oint(\vec{C}\bullet\vec{r})d\vec{l} = \left\{\frac{1}{2}\int_C \vec{r}\times d\vec{l}\right\}\times\vec{C}$$

$$\vec{A}_D(\vec{r}) = \frac{\mu_0}{4\pi r^2}\left[\frac{1}{2}\int_{V'}\vec{r}\,'\times\vec{J}(\vec{r}\,')d\tau'\right]\times\hat{r} \qquad \frac{\mu_0}{4\pi r^2}\left\{\frac{1}{2}\int_C \vec{r}\,'\times Id\vec{l}\,'\right\}\times\hat{r}$$

Now we define magnetic dipole moment in both the cases as

$$\vec{m} = \frac{1}{2}\int_{V'}\vec{r}\,'\times\vec{J}(\vec{r}\,')d\tau' \qquad \vec{m} = \frac{I}{2}\int_C \vec{r}\,'\times d\vec{l}\,'$$

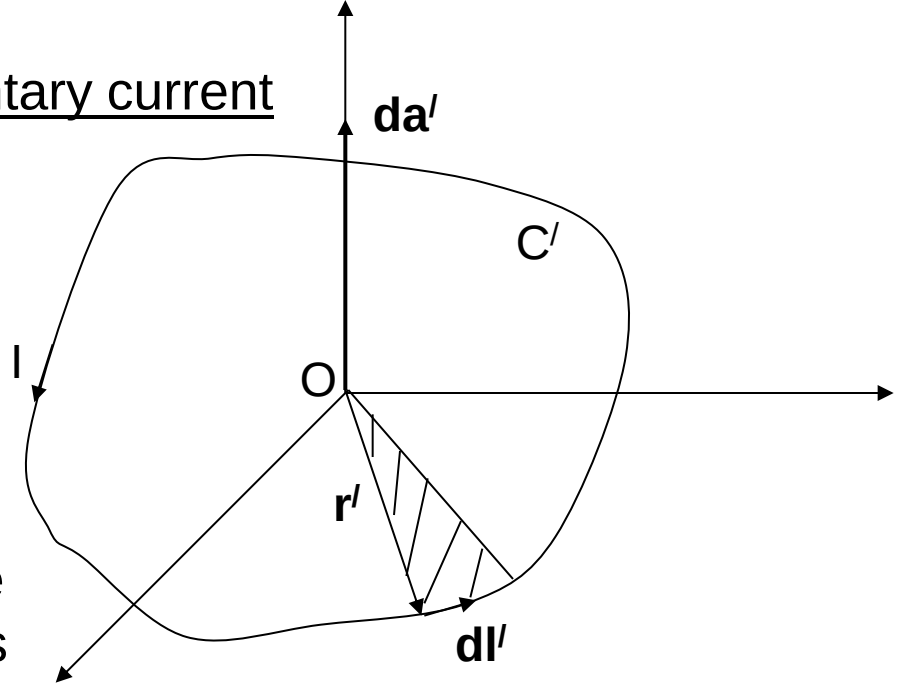
Therefore, vector potential due to magnetic dipole moment is given as

$$\vec{A}_D(\vec{r}) = \frac{\mu_0}{4\pi}\frac{\vec{m}\times\hat{r}}{r^2} = \frac{\mu_0}{4\pi}\frac{\vec{m}\times\vec{r}}{r^3}$$

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}.\hat{r}}{r^2}$$

Dipole moment due to Plane filamentary current

$$\vec{m} = \frac{I}{2}\int_C \vec{r}' \times d\vec{l}'$$



$\vec{r}' \times d\vec{l}'$ **is perpendicular to the plane of the loop**

Its magnitude is equal to the area of the Parallelogram with **r/** and **dl/** as its sides

$\frac{1}{2}\vec{r}' \times d\vec{l}'$ **is then the area of the shaded region**

$$\frac{1}{2}\oint_{C'} \vec{r}' \times d\vec{l}' = \int d\vec{a}' = \vec{S} = S\hat{n}$$

$$\vec{m} = I\vec{S} = IS\hat{n}$$

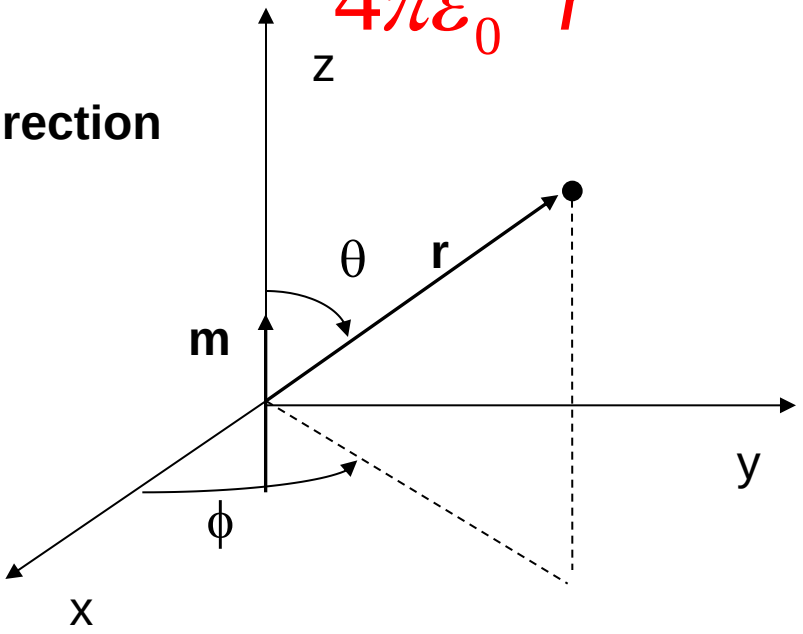
$$\vec{p} = q\vec{d} = qd\hat{d}$$

Magnetic dipole field – field due to magnetic dipole

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi}\frac{\vec{m}\times\hat{r}}{r^2}$$

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}.\hat{r}}{r^2}$$

Put m at the origin and let it point in the z-direction



$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi}\frac{m\sin\theta}{r^2}\hat{e}_\phi$$

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}$$

Magnetic field is

$$\vec{B}_{dip}(\vec{r}) = \vec{\nabla}\times\vec{A} = \frac{\mu_0 m}{4\pi r^3}\left(2\cos\theta\hat{e}_r + \sin\theta\hat{e}_\theta\right)$$

$$\text{Field due to electric dipole} = \frac{p}{4\pi\varepsilon_0 r^3}\left(2\cos\theta\hat{e}_r + \sin\theta\hat{e}_\theta\right)$$

Magnetic field is

$$\vec{B}_{dip}(\vec{r}) = \vec{\nabla} \times \vec{A} = \frac{\mu_0 m}{4\pi r^3}\left(2\cos\theta \hat{e}_r + \sin\theta \hat{e}_\theta\right)$$

$$\text{Field due to electric dipole} = \frac{p}{4\pi\varepsilon_0 r^3}\left(2\cos\theta \hat{e}_r + \sin\theta \hat{e}_\theta\right)$$

In general

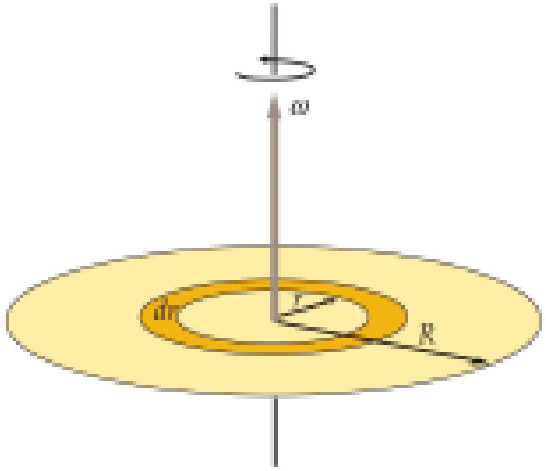
$$B_{dip}\left(\vec{r}\right) = \frac{\mu_0}{4\pi}\frac{1}{r^3}[3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}]$$

$$E_{dip}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{1}{r^3}\left[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}\right]$$

Magnetic Moment of a Rotating Disk

Consider a nonconducting disk of radius R with a uniform surface charge density σ. The disk rotates with angular velocity $\vec{\omega}$.

Calculation of the magnetic moment $\vec{\mu}$:



- Total charge $Q = \sigma(\pi R^2)$.
- Divide the disk into concentric rings of width dr.
- Period of rotation: $T = \dfrac{2\pi}{\omega}$.
- Current within ring: $dI = \dfrac{dQ}{T} = \sigma(2\pi r dr)\dfrac{\omega}{2\pi} = \sigma\omega r dr$.
- Magnetic moment of ring: $d\mu = dI\,(\pi r^2) = \pi\sigma\omega r^3 dr$.
- Magnetic moment of disk: $\mu = \displaystyle\int_0^R \pi\sigma\omega r^3 dr = \frac{\pi}{4}\sigma R^4\omega$.
- Vector relation: $\vec{\mu} = \dfrac{\pi}{4}\sigma R^4\vec{\omega} = \dfrac{1}{4}QR^2\vec{\omega}$.